

# Comparison of Median Regression Tree and Weibull Models for Censored Survival Data

Peter Wang

## Abstract:

This project compares a median regression tree with a Weibull regression model for analyzing censored survival data. The median regression tree is constructed using a residual-based splitting procedure and predicts survival times for different subgroups, adopted from Cho and Hong (2008). The Weibull model is used as a baseline. Model performance is evaluated using different tests to examine the benefits of the Tree model.

## 1. Introduction:

Survival analysis is about the modeling of time-to-event data. The outcome of interest is the time until an event occurs. Examples of event are failure, death, or disease onset. Notice that the outcome is not just whether an event occurs, but when it occurs. However, in many studies, the exact event time is not observed. There may be subjects who leave the study early and subjects who do not experience the event before the study ends. Not observing the actual time to event is called censoring, which makes standard regression methods unsuitable. We want to estimate relative effects of risk factors and predicting survival probabilities/survival times on this type of data.

In this project, we use a Weibull regression model as the baseline method, and we will compare it to the median regression tree method. The Weibull model directly relates covariates to time-to-event outcomes. The model predicts the estimate of survival time. The Weibull model imposes a specific functional form on the data, which can be restrictive. If the true relationship is nonlinear or varies across subgroups, the model may fail to capture the correct relation. As a result, the Weibull model provides a useful and interpretable baseline, but it may not give the most accurate prediction.

The median regression tree is a method that combines the advantages of a median linear regression model and a tree-structured model. Median linear regression model estimates the conditional median of survival time given covariates. The interpretation of estimate is simple: Half of the observations lie above the estimate, and half lie below. This makes the method robust to skewness and outliers. However, the median linear regression model assumes a linear relationship between covariates and the outcome, which is often not the case, especially in clinical data where interactions and nonlinear effects are common. The median linear regression is also difficult to compute which we will cover later.

Tree-structured model can help median linear regression model to resolve the problem of linear assumption. A tree-structured model partitions the covariate space into subgroups and fit simple models within each subgroup. This allows the model to capture nonlinear patterns and interactions without requiring to make distribution assumptions. The resulting structure is easy to interpret. Each terminal node corresponds to a subgroup with a similar set of covariates. However, standard regression trees do not handle censoring well, which motivates us to combine median linear regression in a tree-structured model.

The median regression tree proposed by Cho and Hong in 2008. It combines median regression with a tree-structured model for censored time-to-event data. Each terminal node produces a predicted median

survival time. The method also introduces a residual-based splitting procedure to reduce bias in variable selection.

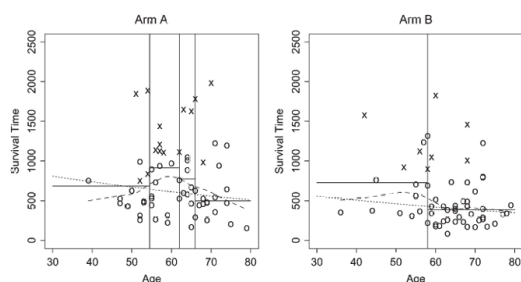


Figure 1. (adapted from Cho and Hong, 2008)

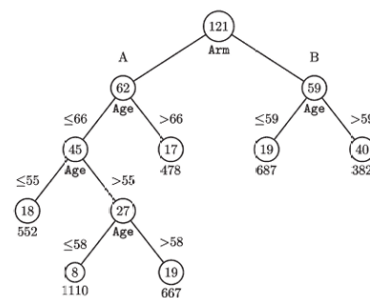


Figure 2. (adapted from Cho and Hong, 2008)

We illustrate the median regression tree using a small-cell lung cancer dataset, where the goal is to study how survival time relates to treatment (Arm A vs Arm B) and patient age. Figure 1 shows the observed time-to-event data, with survival time plotted against age for each treatment group. The pattern is not linear and changes across age ranges. Each vertical line represent a split in the covariate, in each section, we use the median of all data points as our model, represented by the horizontal line, while the dotted line represents the overall median linear regression model. We see that the overall median linear regression model does not present the non-linear relation of the data well. Figure 2 shows the structure of the median regression tree by first splitting on treatment, and then further partitioning each group based on age. Each terminal node corresponds to a subgroup with a predicted median survival time. We note that the median regression tree model represents the data as a set of simple piecewise patterns rather than a single global relationship.

In this report, we will be analyzing the rat tumor dataset. The rat tumor dataset is obtained from the survival package in R, originally reported in Mantel et al. (1977). The median regression tree is well suited for this project because it is designed for censored time-to-event data and allows for direct prediction of survival time while capturing nonlinear effects and subgroup structure. The rat tumor dataset provides an appropriate setting, as it includes censored outcomes along with covariates such as treatment and sex that may interact in non-linear ways. We will compare this method with a Weibull regression model, which serves as a baseline.

## 2. Methods

### 2.1 Median Regression for Censored Data

We begin with median linear regression as the base model.

We model the conditional median of the time-to-event as  $\theta_i = X_i^T \beta$ . The median regression model can then be written as

$$Z_i = \theta_i + \varepsilon_i,$$

where the error terms  $\varepsilon_i$  are independent and identically distributed with an unspecified distribution. This avoids assuming a specific error distribution while still allowing us to model survival time directly.

We first introduce the estimation of the median regression model for uncensored data.

Let  $Z_i$  denote the true time-to-event and  $X_i$  the covariate vector for the  $i$ -th subject. When there is no censoring, median regression estimates  $\beta$  by minimizing the sum of absolute errors:

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n |Z_i - X_i^T \beta|$$

This formulation estimates the conditional median of the response. It does not rely on squared loss, so it is less sensitive to extreme outliers and skewed distributions.

For censored data,  $Z_i$  is not always observed. Instead, we observe  $Y_i = \min(Z_i, C_i)$  and the indicator  $\Delta_i = I(Z_i \leq C_i)$ . Direct minimization is no longer feasible. Following Cho and Hong (2008), we use a censoring-adjusted estimating equation:

$$\sum_{i=1}^n X_i \left[ \frac{I(Y_i \geq X_i^T \beta)}{\hat{G}(X_i^T \beta)} - \frac{1}{2} \right] \approx 0,$$

where  $\hat{G}$  is the Kaplan–Meier estimate of the censoring survival function. A survival function describes the probability that a time-to-event variable exceeds a given time  $t$ , that is,  $S(t) = P(T > t)$ . In this context,  $G(t)$  denotes the survival function of the censoring time  $C$ , so  $G(t) = P(C > t)$ . The Kaplan–Meier estimator is used to estimate this function from the observed data while accounting for censoring. The Kaplan–Meier estimator is a nonparametric estimator of the survival function, introduced by Kaplan and Meier (1958). It estimates the probability that a time-to-event variable exceeds a given time by combining observed event times while properly accounting for censoring. In conclusion, this adjustment corrects for the bias introduced by censoring.

However, median regression has several practical weaknesses. First, the estimating equation is difficult to solve. It is discontinuous, so standard optimization methods such as Newton–Raphson cannot be applied directly. Second, the model assumes a linear relationship between covariates and survival time, which may not hold in practice. Third, categorical variables with many levels require the use of multiple dummy variables, which makes the model more cumbersome to implement. These problems motivate the median regression tree model by Cho and Hong.

## 2.2 Median Regression Tree

The median regression tree replaces one global model with a simple model in each split. Each split divides the data into smaller groups, and each terminal node gives a predicted median survival time for that subgroup. To see the prediction of a subject, we simply look at the terminal node's prediction value following the subject's covariates.

The original paper proposes a full tree-growing procedure followed by cost-complexity pruning to determine the optimal tree size. In this project, we do not implement the pruning step. Instead, we adopt a simpler stopping rule based on statistical significance when selecting splits. This choice keeps the tree close to its maximum potential and makes the implementation much easier, allowing us to focus on the core idea of residual-based partitioning.

Our implementation proceeds node by node. At each node, we first compute a predicted median survival time by taking the median of the observed times within that node. At each node, the paper defines residuals from a fitted median constant model as

$$e_i = \frac{I(Y_i - \hat{\theta} \geq 0)}{\hat{G}(\hat{\theta})} - \frac{1}{2}$$

where  $Y_i$  is the observed time and  $\hat{G}$  is the Kaplan–Meier estimate of the censoring survival function. If the fitted node model is adequate, these residuals should not show any systematic pattern across the covariates. If such a pattern appears, then the node still contains heterogeneity and should be split further. This residual-based approach is one of the main features of the method.

We test the residuals with a curvature test, under the **null hypothesis**: residuals are independent of the covariate (no pattern / no curvature). This is also known as Pearson  $\chi^2$  test for a  $2 \times c$  contingency table constructed by the signs of the residuals and the sample medians (or the categories) of a pair of covariates. Testing the null hypothesis is equivalent to testing whether the rows and columns of the contingency table are independent.

For numerical covariate, we construct the following contingency table (Figure 3), with, where the horizontal axis is the quartile that the data point is in, and the vertical axis is whether the residual of that data point is positive or negative. We notice that the sum for each row and column is constant, we can therefore test if the rows and columns are independent with a  $\chi^2$  test.

|                   | Quartile 1 | Quartile 2 | Quartile 3 | Quartile 4 |
|-------------------|------------|------------|------------|------------|
| Positive residual |            |            |            |            |
| Negative residual |            |            |            |            |

Figure 3.

For categorical data (using gender example), we put each category in the horizontal axis (Figure 4). Similar to the table above, we test whether the rows and columns are independent with a  $\chi^2$  test.

|                   | Male | Female |
|-------------------|------|--------|
| Positive residual |      |        |
| Negative residual |      |        |

Figure 4.

After the test is done for all covariates, we find the covariate with the smallest p-value, being the covariate we want to split first.

After the split variable is selected, we still need to decide where to split it. Following the paper, the split point for a covariate is chosen from the observed values of that covariate so that the two child nodes are as different as possible with respect to the fitted median structure. We test all split candidate under the **null hypothesis**: Residuals are independent of the split.

For numerical covariates  $X$ , we split by:

$$X \leq c \text{ (left split) vs } X > c \text{ (right split)}$$

For categorical covariates  $X$ , we split by:

$$X \in A \text{ (left split) vs } X \notin A \text{ (right split)}$$

where  $A$  is a set of categories of the covariate.

We test all split point  $c$  for numerical covariates or all set of categories  $A$  for categorical covariates, by building a contingency table and test if the rows and columns are independent of the columns (Figure 5). Since the sum of each row and column is constant, we use the  $\chi^2$  test to test if the rows and columns are independent, which is equivalent to testing the null hypothesis.

|             | Positive | Negative |
|-------------|----------|----------|
| Left split  | a        | c        |
| Right split | b        | d        |

Figure 5.

Once a split variable and a split point is chosen, we divide the node into two child nodes and repeat the same procedure within each child node. We stop the recursion when the remaining covariates no longer show clear association with the residual signs. This simplified procedure does not reproduce every technical detail of Cho and Hong (2008), but it preserves the main idea of recursive partitioning and yields an interpretable tree for censored survival data.

### 2.3 Baseline Model: Weibull Regression

To evaluate the strengths and weaknesses of the median regression tree, we compare it with a Weibull regression model as a parametric baseline. We choose the Weibull model because it models time-to-event outcomes directly and was also shown in assignment 4 that it is the best model for the rat tumor data. The Weibull model allows the hazard rate to vary with time, so it is more flexible while keeping a simple parametric form.

Let  $T_i$  denote the survival time for subject  $i$ , and let  $X_i$  denote the covariate vector. Under a Weibull model, the survival time is assumed to follow a Weibull distribution with survival function

$$S(t) = \exp \left\{ - \left( \frac{t}{\rho} \right)^\kappa \right\}, t > 0$$

where  $\rho > 0$  is a scale parameter and  $\kappa > 0$  is a shape parameter. The corresponding hazard function is

$$h(t) = \frac{\kappa}{\rho} \left( \frac{t}{\rho} \right)^{\kappa-1}$$

When  $\kappa = 1$ , the hazard is constant and the model reduces to an exponential distribution model. When  $\kappa > 1$ , the hazard increases with time; when  $\kappa < 1$ , it decreases with time.

The Weibull model is fitted as an accelerated failure time model, written as:

$$\log(T_i) = X_i^T \beta + \sigma \varepsilon_i$$

where  $\beta$  is the vector of regression coefficients,  $\sigma$  is a scale parameter, and  $\varepsilon_i$  follows the extreme-value distribution. Under this parameterization, the Weibull shape parameter is  $\kappa = 1/\sigma$ , and the scale

parameter is determined by the linear predictor.

### 3. Data Description, Experimental Design, and Results

#### 3.1 Data Description

We use the rat tumor dataset from the R package survival, originally reported in Mantel et al. (1977). The dataset contains 300 observations and records the time to tumor development or last follow-up for rats from 100 litters, with three rats sampled from each litter. The variables used in this study are time, the observed time-to-event or censoring time; status, the event indicator, where 1 denotes tumor occurrence and 0 denotes censoring; rx, the treatment indicator, with drug and control groups; sex, the sex of the rat; and litter, the litter identifier (Figure 6). This dataset is suitable for this analysis because it contains censored time-to-event outcomes together with sex and treatment covariates, which makes it a natural setting for comparing a tree-based survival method with a standard parametric model.

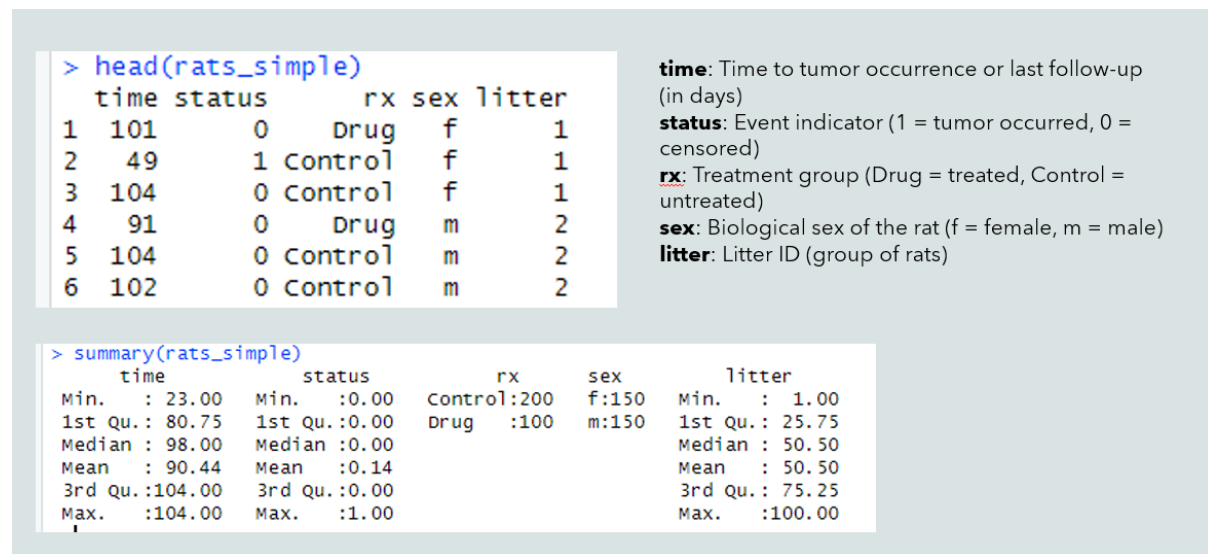


Figure 6.

We have a total of 3 covariates in the model. No additional covariates were created for the tree other than grouping litter into quartiles when carrying out the residual-based split tests, since the numerical split procedure required grouped values in the contingency table as shown in last 2.2. The median regression tree and the Weibull regression model were then compared using the same dataset.

#### 3.2 Experimental Design and Results

We first loaded the rat tumor data from the survival package and kept the variables time, status, rx, sex, and litter. We then recoded rx as a factor with levels “Control” and “Drug” and treated sex as a categorical variable. At the root node, we computed the node median and estimated the censoring survival function, which allowed us to calculate the residuals defined in the paper. We converted these residuals into positive and negative signs and tested their association with each covariate. For rx and sex, we used Fisher’s exact test. For litter, we grouped the values into quartiles and applied a chi-square test. The covariate with the smallest p-value was selected for splitting. We then repeated the same procedure within each child node until no further useful split was found.

At the root node, we computed the node median to be 104. The p-values of the covariates are:

$$p_{rx} = 0.612, p_{sex} = 0.186, p_{litter} = 0.200$$

Notice that at this node, more than half of the data are censored with observed time  $\geq 104$ , so the real median is unknown, and 104 is our closest guess for the median. This is our closest approach to treat this dataset using the paper's method, and we will be discussing the consequences of applying the model on this dataset.

We split among the sex variable because it has the smallest p-value (Figure 7).

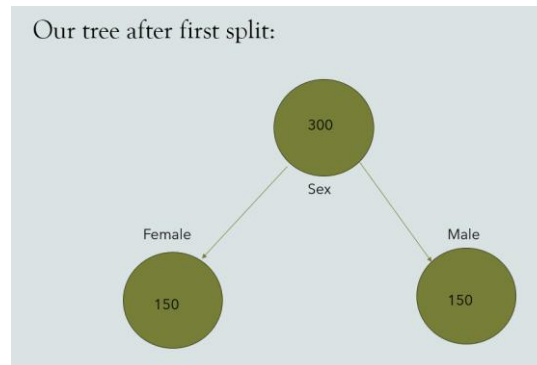


Figure 7.

After splitting by sex, the same procedure is applied within each subgroup.

Within the female group, the following p-values are obtained:

$$p_{rx} = 0.036, p_{litter} = 0.152$$

Since treatment (rx) has the smallest p-value, the female node is further split by treatment.

Within the female control group, only litter is considered for further splitting:

$$p_{litter} = 0.487$$

Within the female drug group:

$$p_{litter} = 0.226$$

Within the male group, the p-values are:

$$p_{rx} = 0.625, p_{litter} = 0.774$$

Since treatment (rx) has a much smaller p-value, the male node is split by treatment.

Within the male control group, only litter is considered for further splitting:

$$p_{litter} = 0.487$$

Within the male drug group:

$$p_{litter} = 0.226$$

After 2 levels of split, we have the following tree (Figure 8):

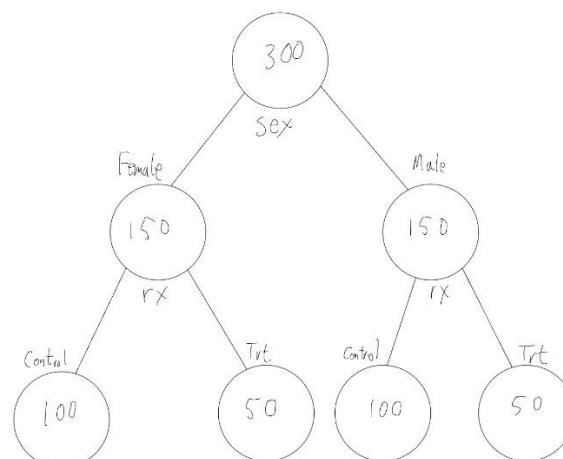


Figure 8.

This is going to be our final tree structure, and we will show why.

The final tree structure is:

- Root split: sex
- Female branch split: rx
- Male branch split: rx

The predicted survival times at the terminal nodes are:

- Female, Control: 102
- Female, Drug: 103
- Male, Control: 104
- Male, Drug: 104

Notice that both sub-nodes for male are approximated median because more than half of the data are censored with observed time  $\geq 104$ .

We previously see that the male node has extremely high p-values for both covariates. This is good evidence that the split is not going to be effective on improving estimation quality. We also see from the result after splitting, we get the same prediction value of 104 for both sub-nodes for male. This proves that the split was ineffective, so we should stop splitting any further; in fact, we should have stopped before this.

We also stop splitting the sub-nodes of female for the following reasons:

1. High p-value, meaning that residual and split of litter is more likely independent
2. Litter is not a significant covariate according to the Weibull model (we will build next)
3. To keep the model simple and balanced (since we stopped splitting male)

Our fitted Weibull regression model is as the follows (Figure 9):

```
> weib_model_all <- survreg(Surv(time, status) ~ rx + sex + litter,
+ data = rats_simple,
+ dist = "weibull")
> summary(weib_model_all)

Call:
survreg(formula = Surv(time, status) ~ rx + sex + litter, data = rats_simple,
dist = "weibull")

              value Std. Error      z      p
(Intercept)  5.08975    0.12138  41.93 < 2e-16
rxDrug       -0.21502    0.08605  -2.50 0.01246
sexm         0.82051    0.22233   3.69 0.00022
litter       -0.00224    0.00144  -1.56 0.11883
Log(scale)  -1.32993    0.14033  -9.48 < 2e-16

Scale= 0.264

weibull distribution
Loglik(model)= -260.3   Loglik(intercept only)= -287.1
    Chisq= 53.65 on 3 degrees of freedom, p= 1.3e-11
Number of Newton-Raphson Iterations: 9
n= 300
```

Figure 9.

For predictive evaluation, we randomly split the data into 70% training data and 30% test data, fit the Median Tree model and the Weibull model on the training set, and generated predicted median survival times for both training and test observations. To compare predictive performance across methods on a common scale, we used the mean absolute error (MAE),

$$\frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

which measures the average absolute difference between the observed time and the predicted survival

time. This metric is applicable for both models and allows comparison between them.

Mean absolute error using all observations:

- Tree (Train): 13.60
- Tree (Test): 13.06
- Weibull (Train): 101.99
- Weibull (Test): 97.43

The regression tree achieves significantly lower prediction error than the Weibull model on both the training and test datasets. In particular, the tree model reduces the prediction error by more than 80 days on average.

Mean absolute error using only uncensored observations:

- Tree (Train): 26.26
- Tree (Test): 23.64
- Weibull (Train): 48.79
- Weibull (Test): 41.26

When restricting to uncensored observations, the prediction error decreases for the Weibull model. This is because the Weibull model predicts censored value way above the observed value. The Tree model has increased prediction error. This is because the Tree model predicts the censored value at its observed value.

We see a graph of the difference (Figure 10).

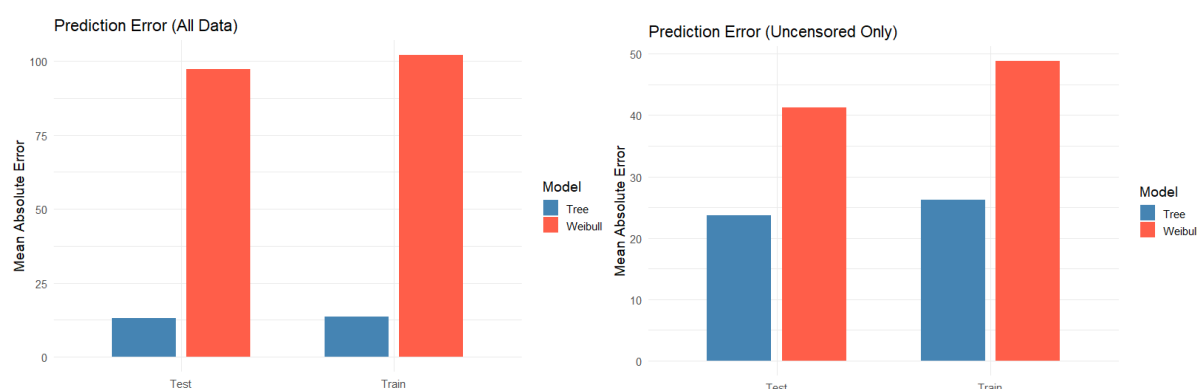


Figure 10.

The regression tree consistently outperforms the Weibull model in terms of MAE. On the full dataset, the difference is particularly large (e.g., 13.06 vs 97.43 on the test set), indicating that the Weibull model produces predictions that are far from the observed values.

When using only uncensored observations, the performance gap becomes smaller (e.g., 23.64 vs 41.26 on the test set), suggesting that part of the Weibull model's poor performance is due to the influence of censored data.

Overall, the tree model provides more accurate predictions of survival time across both training and testing datasets.

In addition to prediction error, model performance is evaluated using the concordance index (C-index), which measures the ability of a model to correctly rank survival times.

The C-index is based on pairwise comparisons between observations. For each comparable pair of

individuals, the model prediction is considered concordant if the individual with the shorter observed survival time is assigned a shorter predicted survival time. Pairs that cannot be ordered due to censoring are excluded from the calculation.

$$C = \frac{\text{Number of concordant pairs}}{\text{Number of comparable pairs}}$$

We only use pairs where event order is known (both uncensored, observed value < censored value). If two objects have same estimates, it's counted as 0.5 concordant pairs.

The C-index is then computed on the test dataset for both models. The results are:

- Weibull model: 0.741
- Tree model: 0.713

These results indicate that the Weibull model provides better ranking of survival times compared to the regression tree.

The higher C-index of the Weibull model suggests that it is more effective at capturing the relative ordering of survival outcomes. This test is not a test designed for the Tree model to perform well, because it is highly likely for the tree model to give the same prediction number for two observations (as long they fall into the same covariates set). Despite this disadvantage, the Tree model's ranking performance is only slightly weaker, suggesting that the Tree model is still a strong model in general.

#### 4. Conclusion

In this project, we compared the median regression tree with the Weibull regression model for analyzing censored survival data using the rat tumor dataset. The two models differ fundamentally in their assumptions and objectives: the Weibull model assumes a parametric structure on the survival distribution, while the median regression tree uses a data-based partitioning approach.

From the prediction error analysis, the regression tree consistently achieved much lower mean absolute error than the Weibull model. Even when restricting the evaluation to uncensored observations, the tree model still outperformed the Weibull model. This test is to demonstrate that the Tree model provides an interpretable structure that captures subgroup effects and nonlinear relationships.

When evaluating ranking performance using the concordance index, the Weibull model performed slightly better. This indicates that the Weibull model is more effective at capturing the relative ordering of survival times. The lower C-index of the tree model is expected, as the tree assigns same predicted values to individuals within the same terminal node, which reduces its ability to distinguish differences in risk.

It is also important to note that the rat tumor dataset contains a large proportion of censored observations at the study endpoint, which makes the true median survival time difficult to estimate. This limitation affects both models, but it particularly influences the tree model, where several node medians are approximated by the maximum observed time. Due to this problem, our split p-value is always high, making our splits not effective, as the prediction numbers are similar in different terminal nodes. This evinces that the Tree model does not perform much better than the Median Linear Regression model without the tree structure. We also notice that the Tree model does not have a numerical covariate that is highly correlated with the residuals or highly significant to the survival time, which means we do not

see the strongest advantage of the tree model in tracking non-linear effects in a numerical covariate. In conclusion, the Tree model would perform a lot stronger in a data set with less censored data, more numerical covariate correlated to the residuals, and numerical covariates that is more significant to the survival time.

#### REFERENCES

1. Cho, H., & Hong, Y. (2008). Tree-structured median regression for censored survival data.
2. Kaplan, E. L., & Meier, P. (19 58). Nonparametric estimation from incomplete observations.
3. Mantel, N., Bohidar, N. R., & Ciminera, J. L. (1977). Rat tumor data study.
4. University of Waterloo, STAT 437 course note
5. 桓峰基因 (2018). Topic 13. 临床预测模型之一致性指数 (C-index), Topic 13. 临床预测模型之一致性指数 (C-index) - 知乎